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Enron Corp.
Case No. EC-2002-01038
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Date: 01/15/2003
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From: Jeff Dasovich <jeff.dasovich@enron.com>
Sent: Fri, 9 Feb 2001 02:46:00 -0800 (PST)
To: Karen Denne <karen.denne@enron.com>
Subject: Re: eCompany Now Brief

My techie pals showed me the feature on Enron in the most recent version of eCompany. What a great feature. Congratulations.

Best,
Jeff

Karen Denne
02/09/2001 09:12 AM

To: Jeff Dasovich/NA/Enron@Enron
cc:
Subject: eCompany Now Brief

This is in today's online "Future Boy" column in eCompany Now -- we had a great meeting with them on Wednesday!

Future Boy

Bringing Opto-Electronics Out of the Stone Age

By Erick Schonfeld at http://www.ecompany.com

Optical networks. It sounds so cutting edge. Billions of bits of data riding on beams of light. The most important, most urgent, information in the world encapsulated within photonic pulses. Yet the components that go into optical gear are often still assembled by hand. Go into a factory where optical equipment is made and you'll find acres of work benches manned by technicians in white lab coats. There is not even an assembly line. It's labor-intensive and practically preindustrial.

If optical networks, and ultimately optical computing, are ever going to rival their electronic predecessors, the companies that manufacture the components that make them possible must first master the art of mass production. As simple as this may sound, it is in fact extremely challenging. Photonic components -- such as

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laser pump diodes, specialty fibers, and optical chips with so-called waveguides that direct light through controlled channels -- are typically made from materials that are not well understood, in contrast to the silicon-based chips used in computers, routers, and switches. "Every industry has to go through an evolution of materials knowledge and an experience curve that just takes time," explains Rick Tompane, CEO of Gemfire, a startup that is figuring out how to make optical-integrated circuits using standard semiconductor equipment.

There's a big difference between making components that are designed to move electrons, and manufacturing components that move pulses of light. An electronic chip, for instance, consists of circuits of metal etched onto silicon. In contrast, an optical chip is usually made from more exotic materials, such as lithium niobate. And instead of metal circuits, it has channels, or waveguides, that guide light passing through it to a desired strand of optical fiber. Rather than etched grooves, these tiny channels are solid veins with a different refractive index than the surrounding material. Learning how to adapt electronic chip-making techniques to produce waveguides and other components in mass quantities has been difficult -- and yet optical-networking companies have to figure this out, if only because the current system is so arduous and antiquated. Many pieces of optical equipment require dozens of laser diodes, which often have to be manually connected to other components. For instance, dense wave division multiplexing (DWDM) boxes, which allow telecom companies to send 32, 40, or even 96 wavelengths of light down a single fiber-optic strand, require different laser diodes that each emit light at slightly different wavelengths. If each of these diodes is inserted by hand, too many opportunities exist for things to go wrong.

But companies such as Gemfire and U.K.-based Southampton Photonics are beginning to automate the manufacture of key optical components. Gemfire has figured out a way to combine diode arrays and waveguides into optical integrated circuits that can be produced using automated equipment. Theoretically, these circuits could be used in DWDM boxes, eliminating the need to have 40 separate laser diodes and to connect each one by hand to a waveguide chip. For now, Gemfire is trying to get these chips into the optical amplifiers that regenerate fading light signals after they've traveled long distances. Gemfire has developed a diode array integrated with eight waveguides that would replace eight separate amplifiers, and thus could amplify eight separate strands

of fiber. Over time, Gemfire will combine even more elements into its optical integrated circuits, including traditional electronic circuitry, so that conversion from the optical to the electrical domain can happen right on the chip. It is this approach that has allowed the private company to attract heavyweight investors such as Cisco Systems, Corning, Intel, and Kleiner Perkins. Floyd Kvamme, a partner from Kleiner who sits on Gemfire's board, was one of the founders of National Semiconductor. "So much of what we are doing reminds him of what they did in the early days of National Semiconductor," says Tompane.

Southampton Photonics offers a similar amplifier product. It, too, has invented an amplifier that can handle eight separate strands, except that its shares diodes between them. But Southampton is also branching out into some more daring pursuits -- as one might expect from a company co-founded by David Payne, who 15 years ago led the team that invented the erbium-doped fiber amplifier, the basis of most of today's optical amplifiers. Payne is now looking into other ways to control and manipulate light. One of the more intriguing solutions is a specialty fiber, known as a Bragg grating, that acts as a filter because its index of refraction changes along its length. Working with these fibers, equipment-makers might no longer have to build DWDM boxes with 40 specialty diodes. Instead, they could simply stock a box with a product from Southampton that combines standard, off-the-shelf diodes with these Bragg gratings, which are stuck in front of the diodes to get whatever wavelengths may be desired. Southampton Photonics makes these fiber filters through an automated, computer-controlled process that can create custom filters for clients in a matter of days rather than weeks. And the process is so fine-tuned that it will allow DWDM manufacturers to reduce the separation between the wavelengths going down a single fiber to 25 GHz, as opposed to the 50 to 100 GHz intervals that are considered state-of-the-art today.